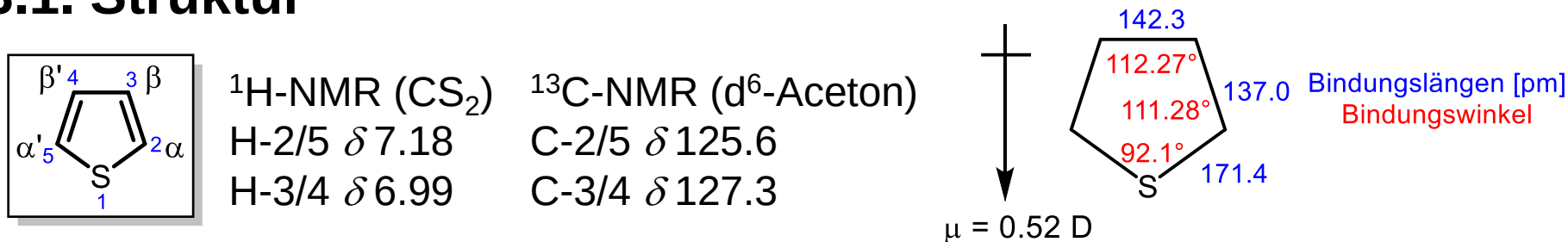


4.3. Thiophen

4.3.1. Struktur



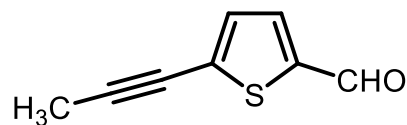
farblose, geruchslose, wasserunlösliche Flüssigkeit (Schmp.: -38°C , Sdp.: 84°C)

Steinkohlenteer, dort: Benzolfraktion (Sdp. Benzol: 80°C)

Empirische Resonanzenergie: $\sim 120 \text{ kJ mol}^{-1}$; Dewar-Resonanzenergie: $\sim 27.2 \text{ kJ mol}^{-1}$

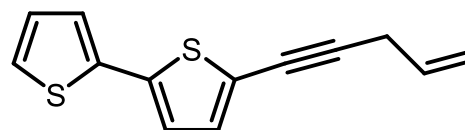
$\Rightarrow$ Aromatizität geringer als die des Benzols, aber größer als die des Furans.

4.3.2. Thiophenhaltige Natur- und Wirkstoffe

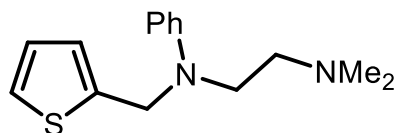


Junipal

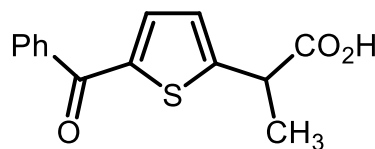
aus *Daedelia juniperina*



aus *Echinops sphareocephalus*



Methaphenilen
(Antihistaminikum)



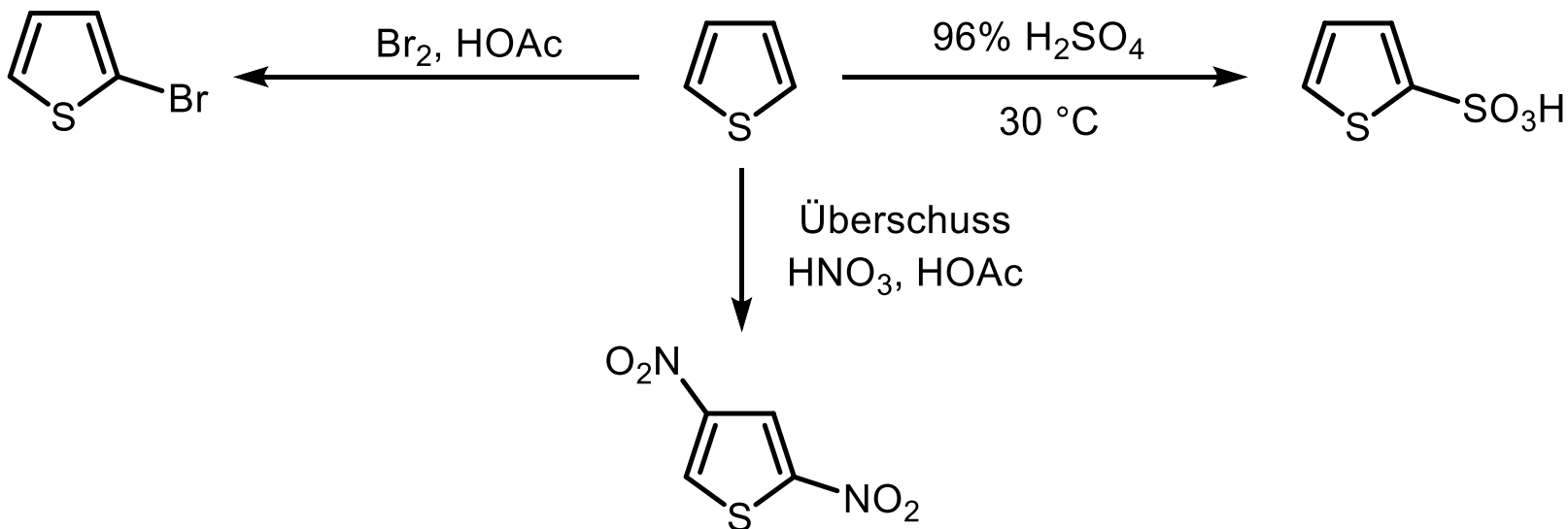
Tiaprofensäure
(Antiphlogistikum)

4.3.3. Thiophen-Synthesen

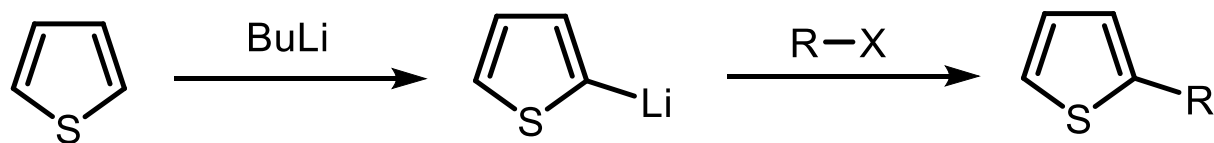
4.3.3.1. Ringtransformationen

Elektrophile Substitutionsreaktionen

44



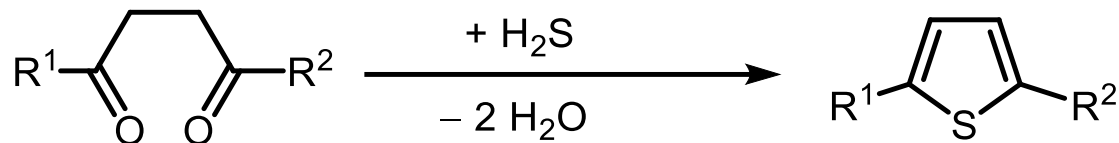
Metallierung



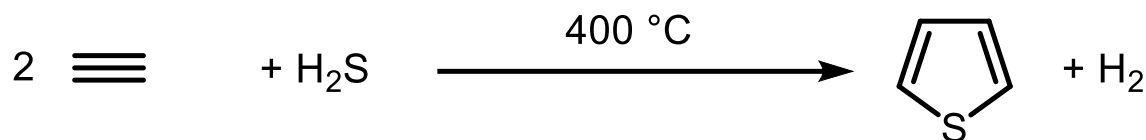
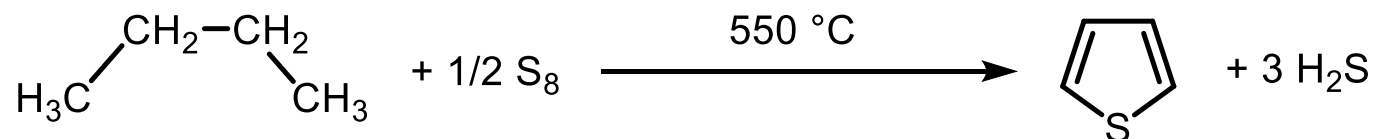
4.3.3.2. Ringaufbauende Synthesen

Paal-Synthese

45

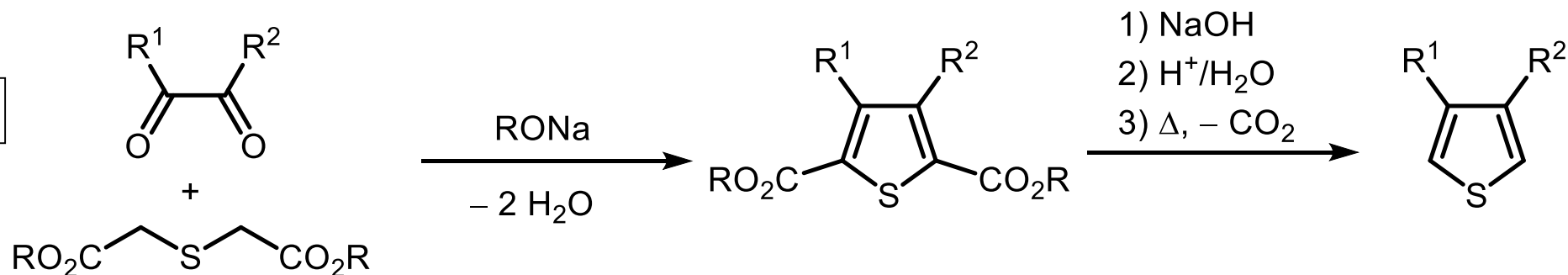


Gasphasen- und Hochdruck-Reaktionen



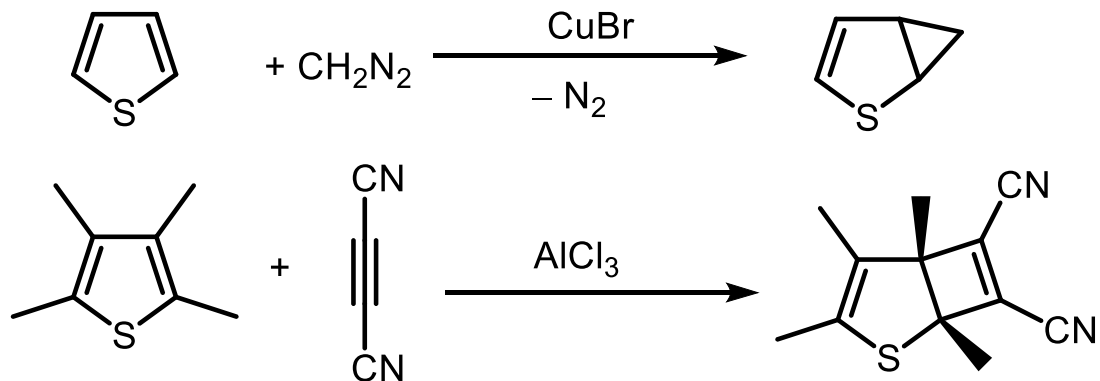
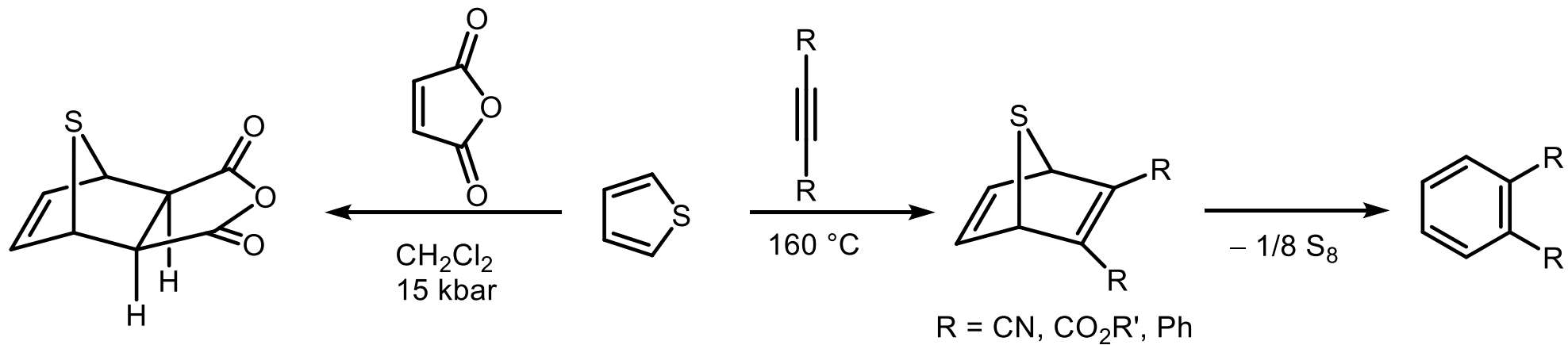
Hinsberg-Synthese

46



4.3.4. Reaktionen

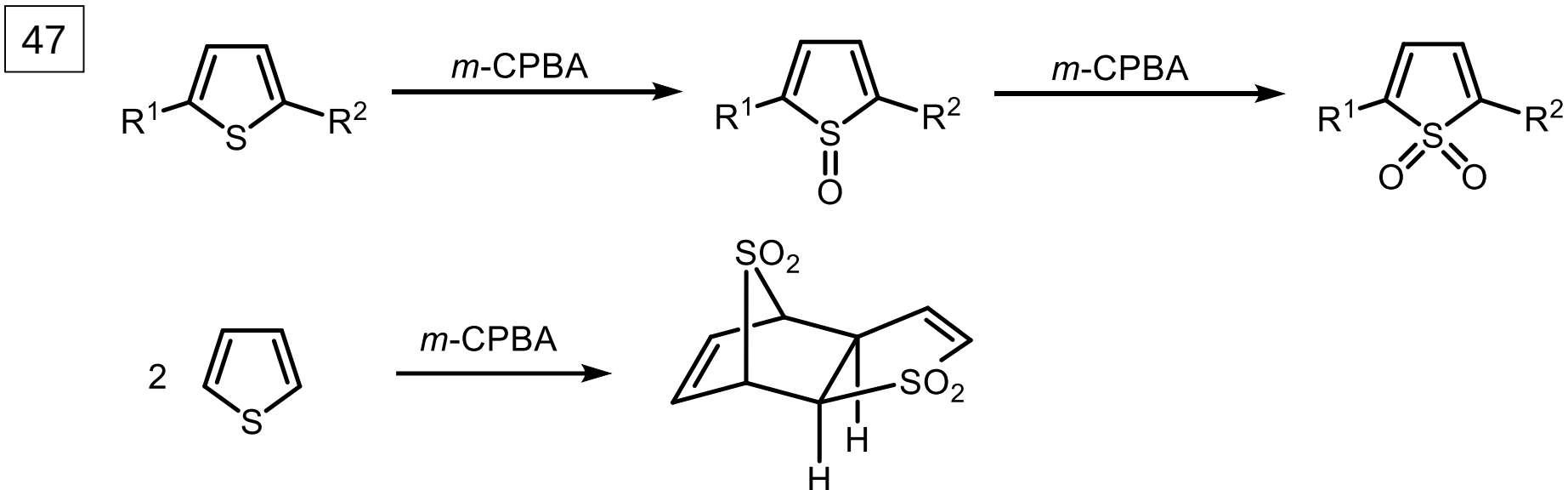
Additionsreaktionen



4.3.4. Reaktionen

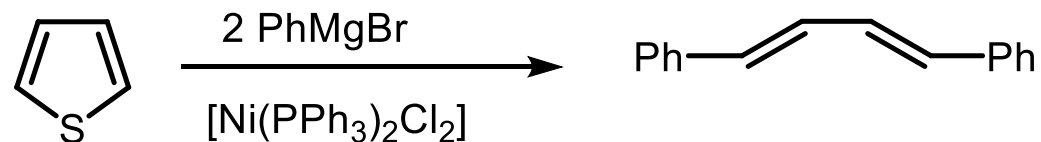
Fortsetzung

Oxidation

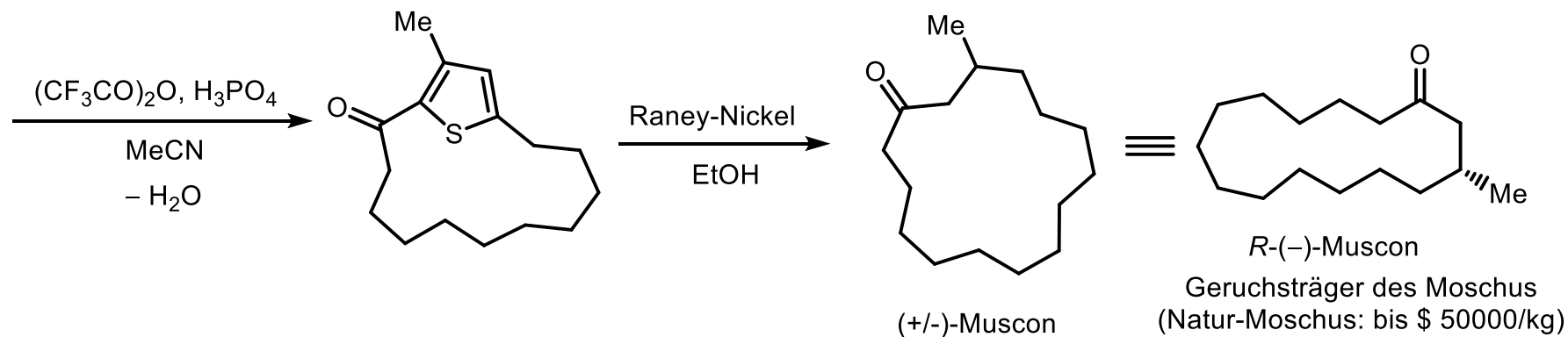
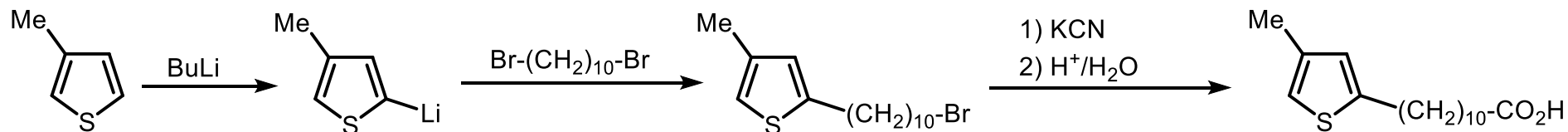
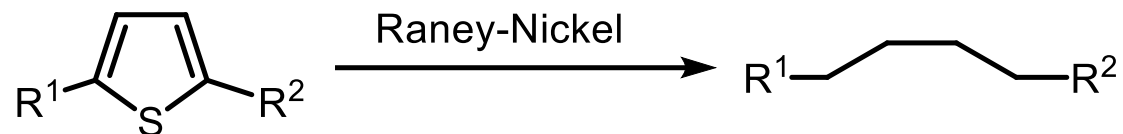


Fortsetzung

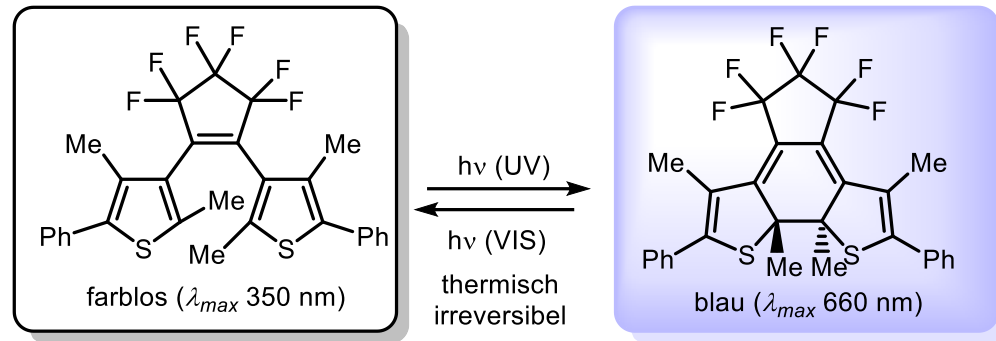
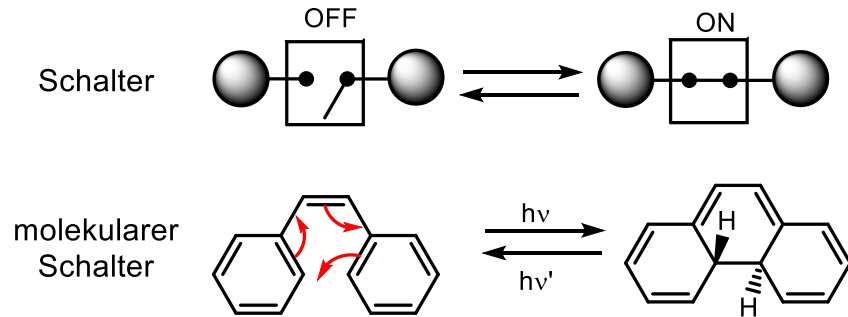
Ringöffnungsreaktionen (reduktive Desulfurierung)



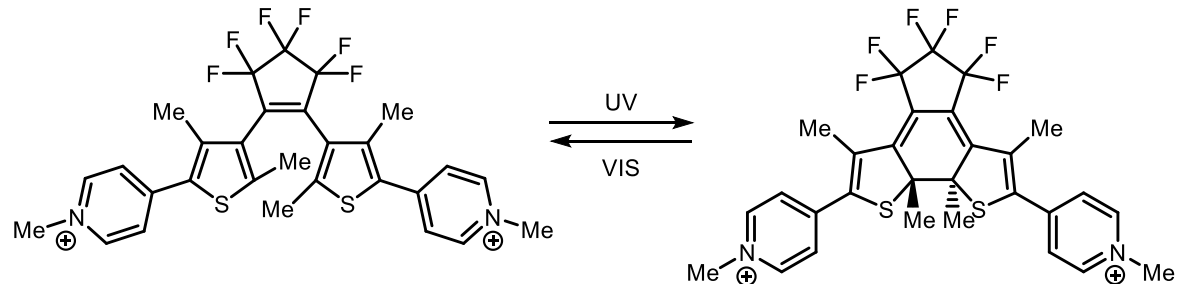
48



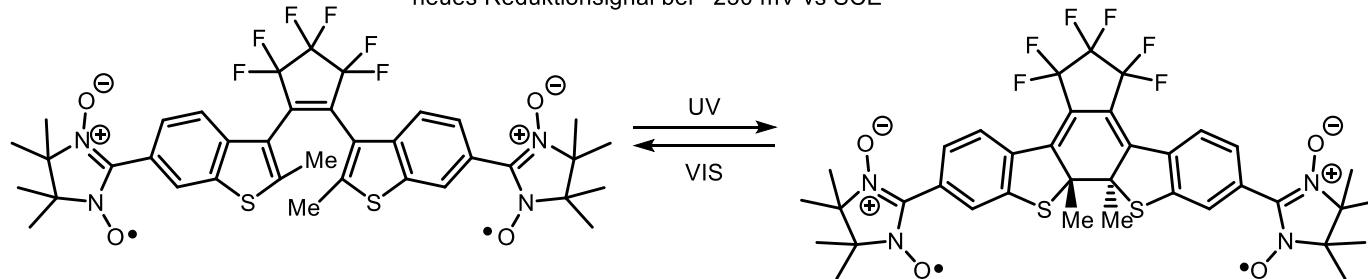
4.3.5. Thiophene in photoschaltbaren Molekülen



M. Irie, *Chem. Rev.* **2000**, 100, 1685



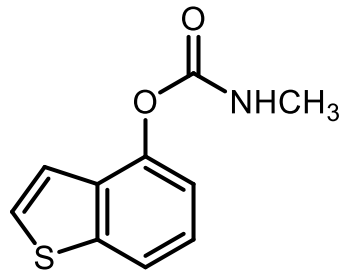
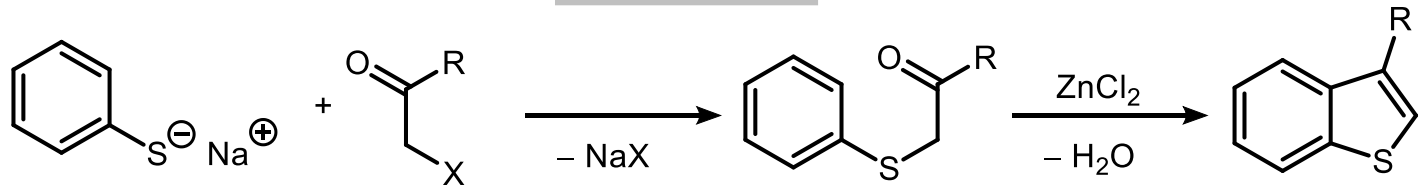
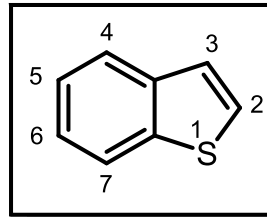
neues Reduktionsignal bei -230 mV vs SCE



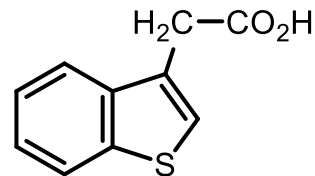
neues antiferromagnetisches Verhalten des Nitronylnitroxid

4.3.6. Benzothiophene

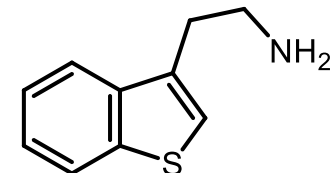
Benzo[*b*]thiophen



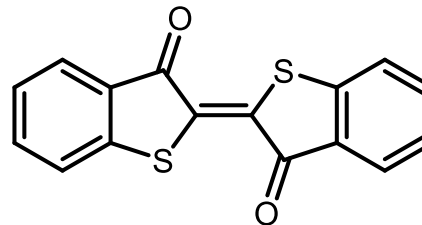
Mobam
(Insektizid)



Benzo[*b*]thiophen-3-ylacetic acid
(Phytohormon)



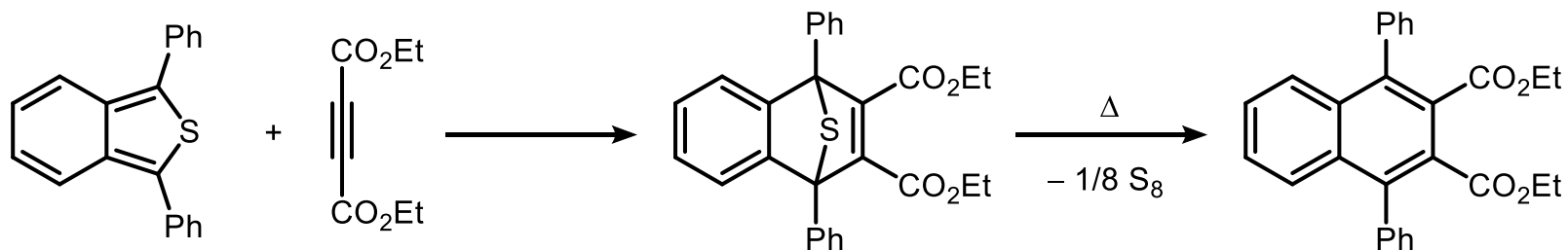
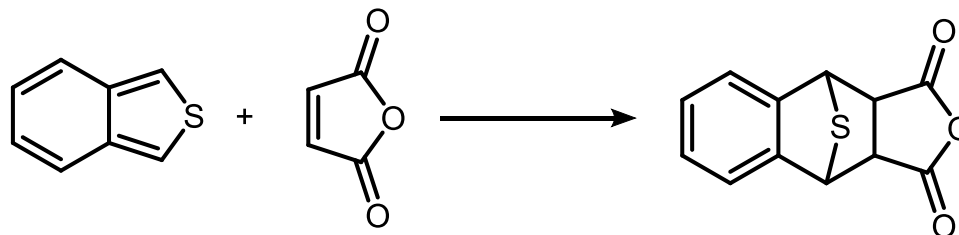
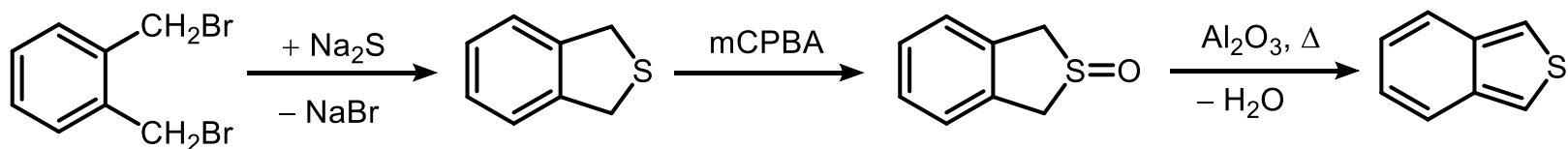
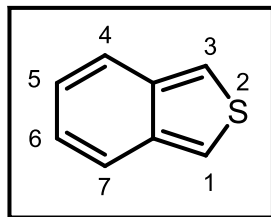
stärkere Wirkung auf das ZNS
als Tryptamin



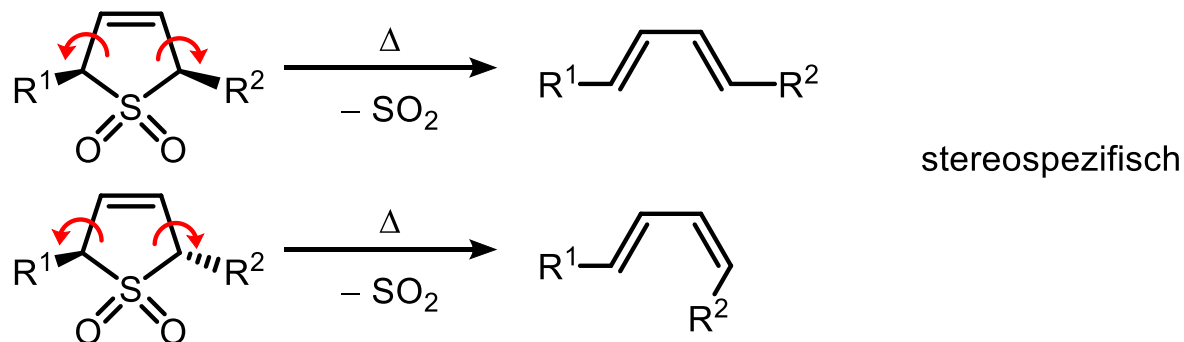
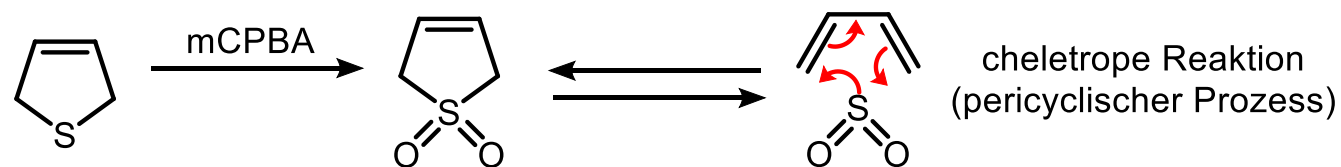
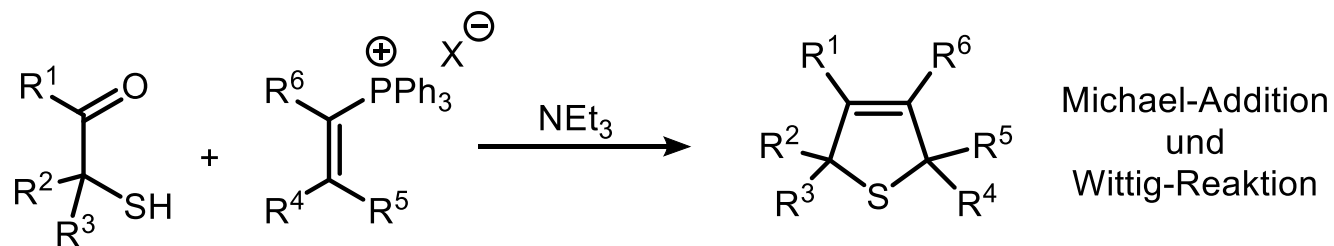
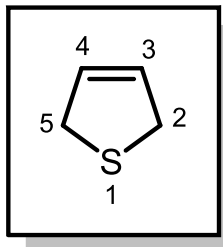
Thioindigo
(rotvioletttes Pigment)

4.3.6. Benzothiophene Fortsetzung

Benzo[c]thiophen

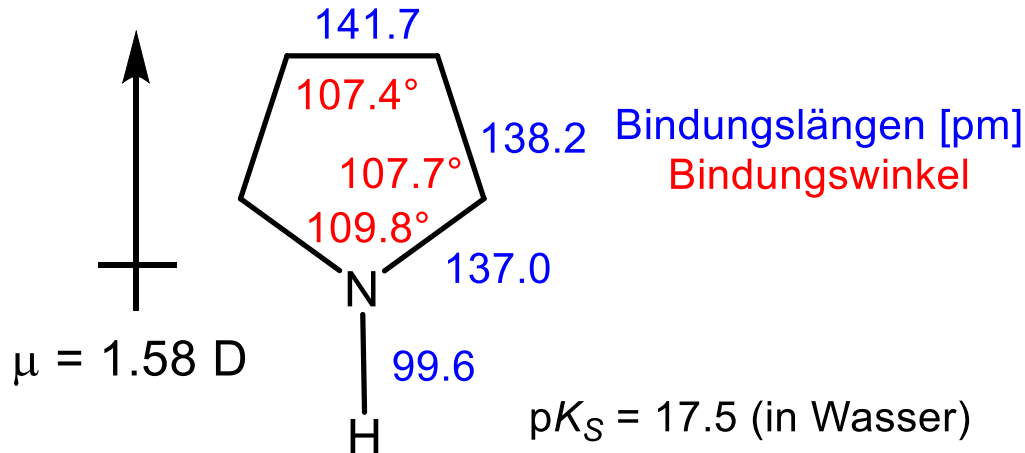
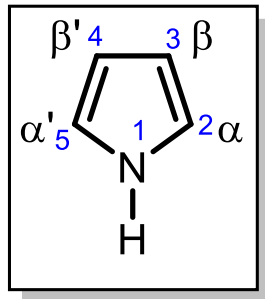


4.3.7. 2,5-Dihydrothiophen

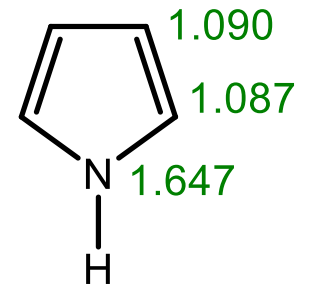


4.4. Pyrrol

4.4.1. Struktur



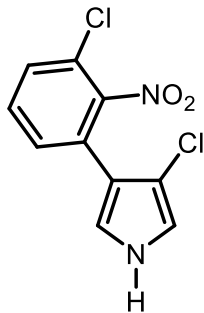
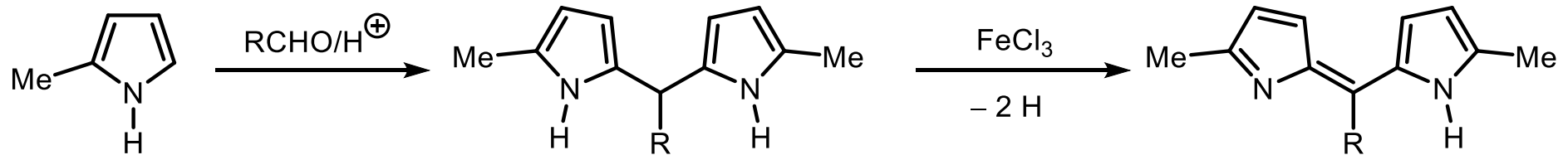
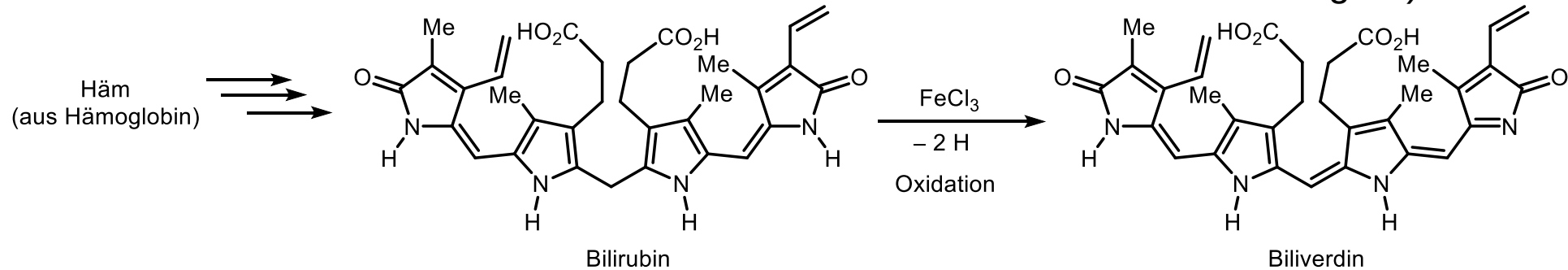
π -Elektronendichte



Heteroaren	Furan	Pyrrol	Thiophen	Benzol
Heteroatom	O	N	S	-
Elektronegativität	3.5	3.0	2.5	(2.5)
Empirische Resonanzenergie [kJ/mol]	80	100	120	150
Dewar-Resonanzenergie [kJ/mol]	19	22	27	95

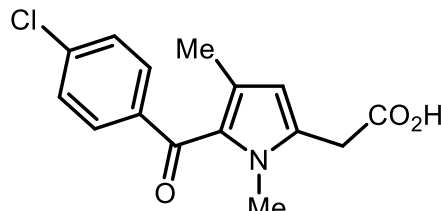
4.4.2. Vorkommen

Bilirubin: Abbauprodukt des Blutfarbstoffs in der Galle (lat. "bilis" Galle; "ruber" rot)



Pyrrolnitrin

(antimykotisch wirksames Antibiotikum) (COX-Inhibitor, nichtsteroidales entzündungshemmendes Medikament mit fiebersenkender Wirkung, seit 1993 vom Markt wegen allergischer Reaktionen)



Zomepirac

4.4.3. Pyrrol-Synthesen

4.4.3.1. Ringtransformationenen

Reaktionen in der α -Position

2-Halopyrrol:

durch Reaktion von Pyrrol mit NCS bzw. NBS;

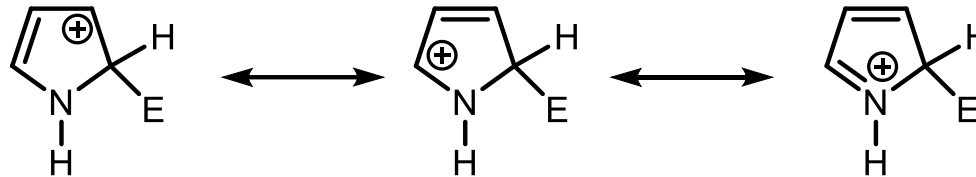
2,3,4,5-Tetrahalopyrrol:

durch Reaktion von Pyrrol mit SO_2Cl_2 , NaOCl bzw. Br_2 ;

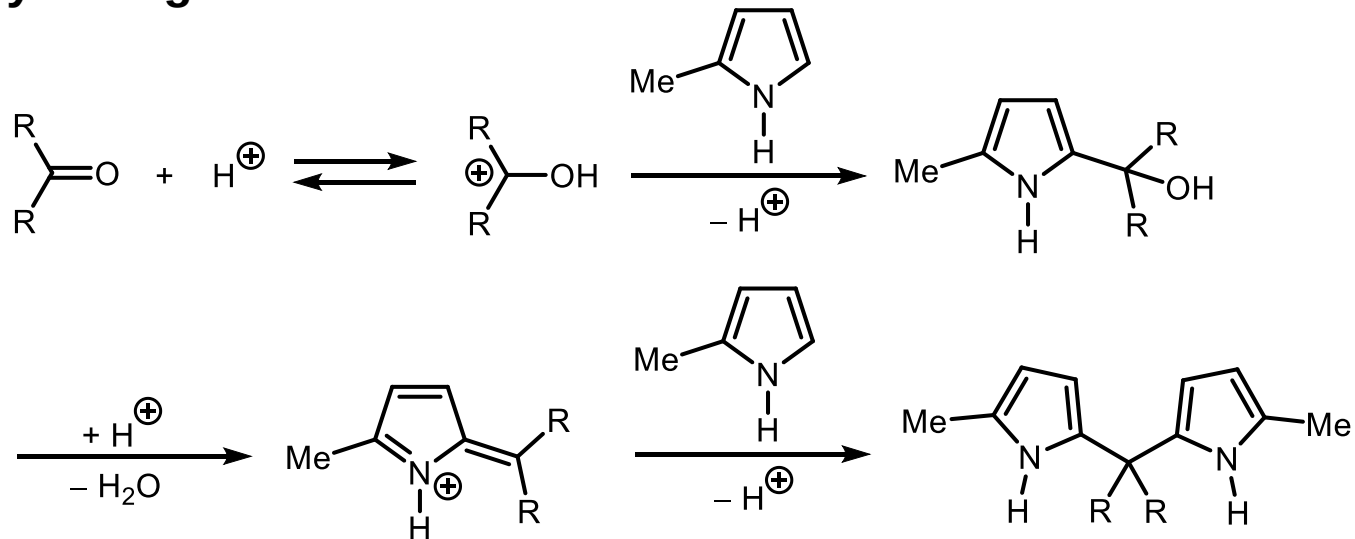
2-Nitropyrrol:

durch Reaktion von Pyrrol mit $\text{HNO}_3/\text{Ac}_2\text{O}$ bei -10°C ;

50



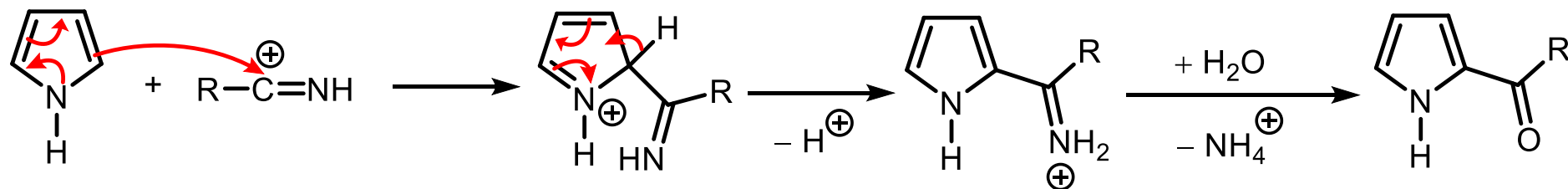
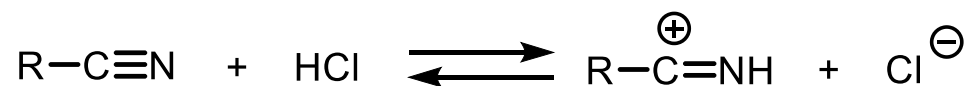
Hydroxyalkylierung



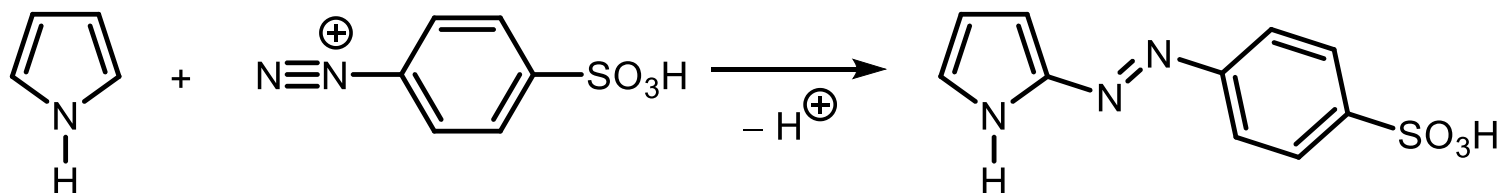
4.4.3.1. Ringtransformationen

Fortsetzung

Houben-Hoesch-Acylierung



Azokupplung (ebenso: Formylierung, Acylierung)

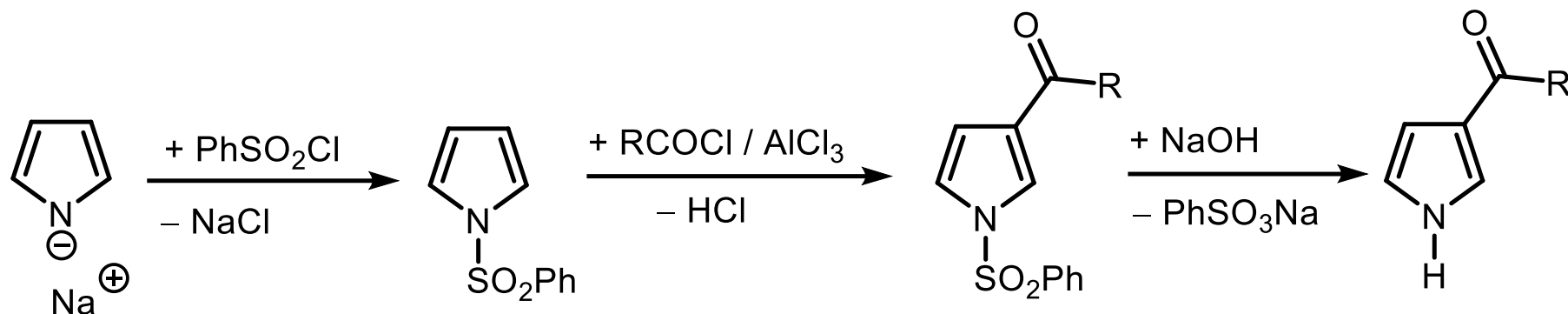


4.4.3.1. Ringtransformationen

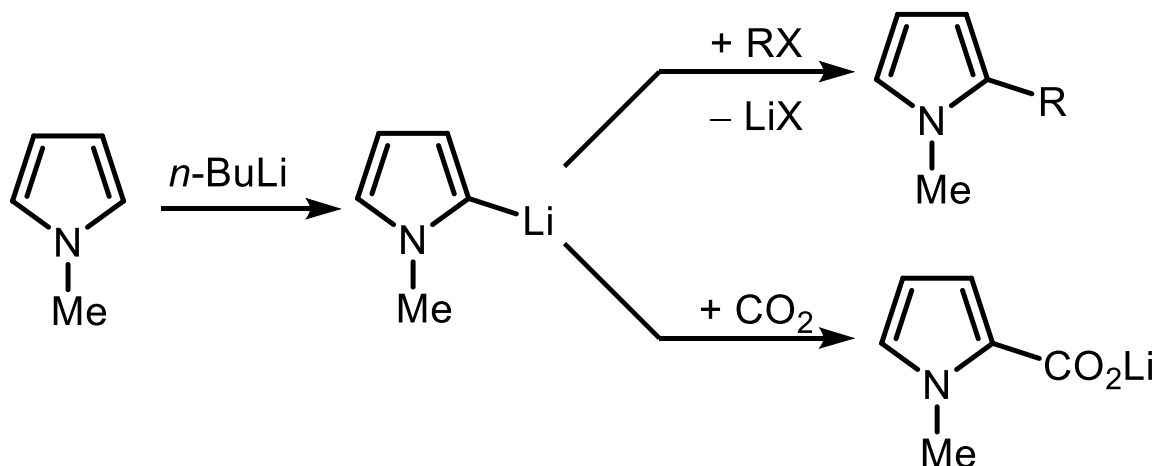
Fortsetzung

Acylierung der 3-Position: vorher *N*-Sulfonylieren (1. NaH, 2. PhSO₂Cl, 3. RCOCl/(AlCl₃), 4. NaOH).

51

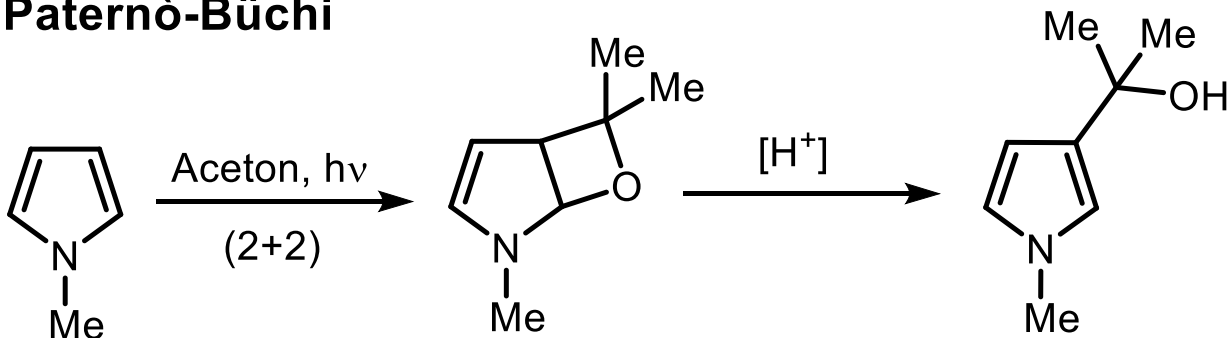


Lithiierung der 2-Position: vorher *N*-Methylieren (1. NaH, 2. MeI, 3. BuLi, 4. z. B. CO₂ zum Li-Salz).

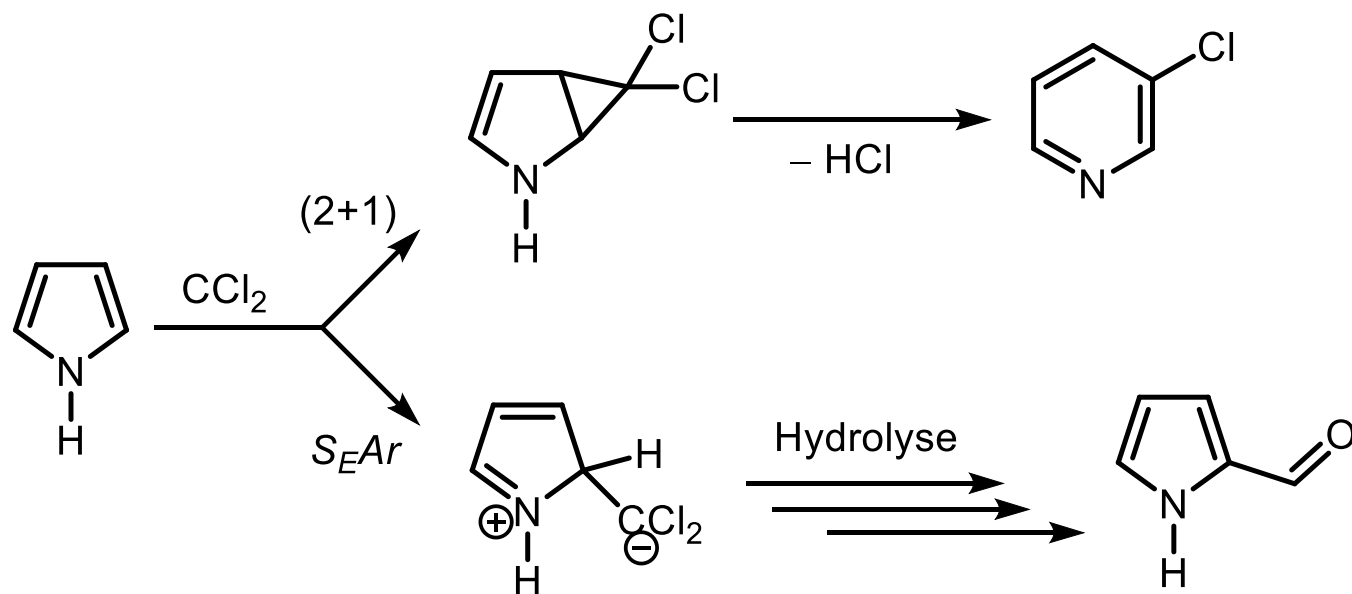


Cycloadditionen

Paternò-Büchi



Cheletrope (2+1)-Cycloaddition



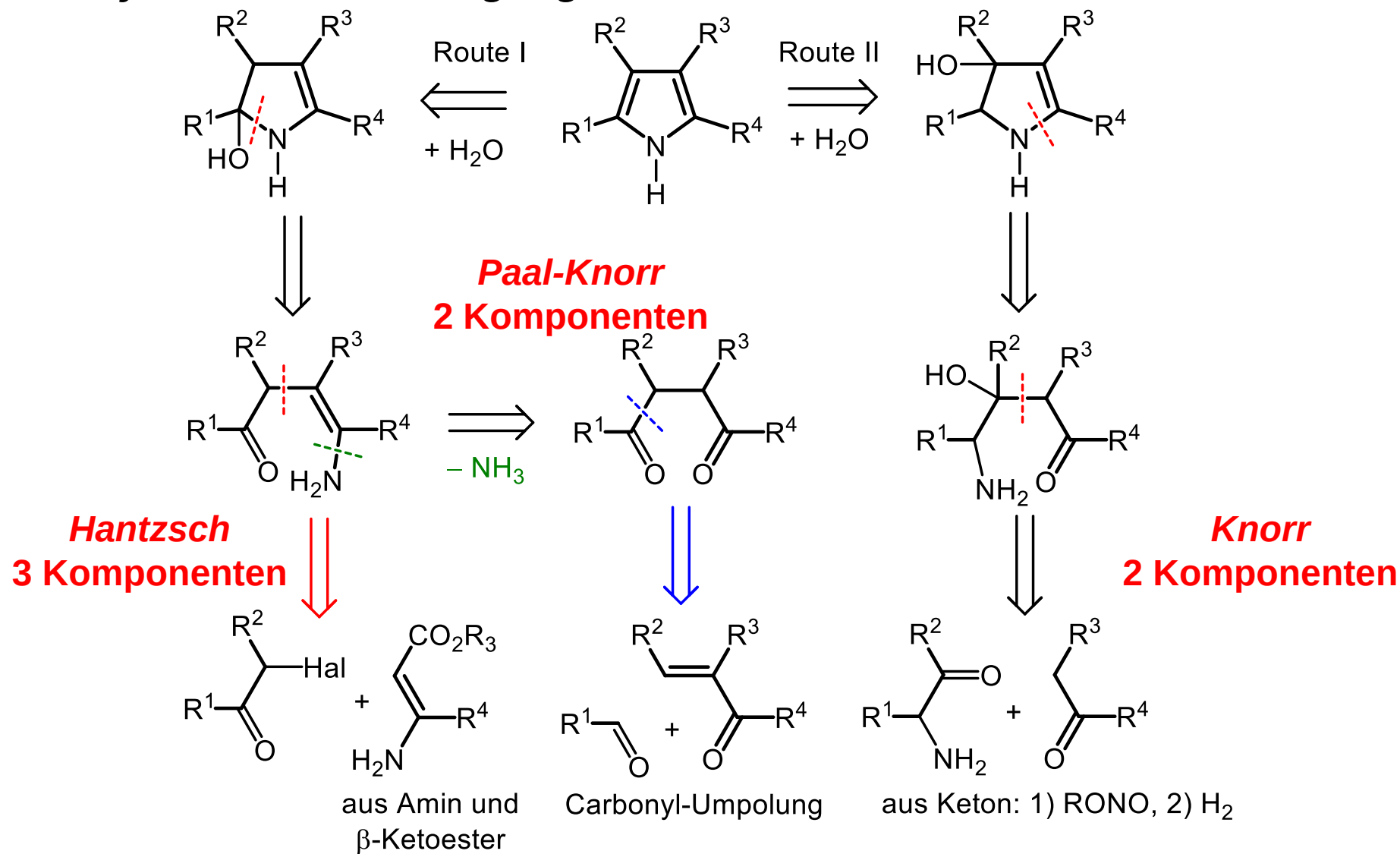
(2+1)-
Cycloaddition,
wenn CCl_2 aus
 NaO_2CCl_3

Reimer-
Tiemann,
wenn CCl_2 aus
 $CHCl_3/NaOH$

Reimer-Tiemann-Synthese

4.4.3.2. Ringaufbauende Synthesen

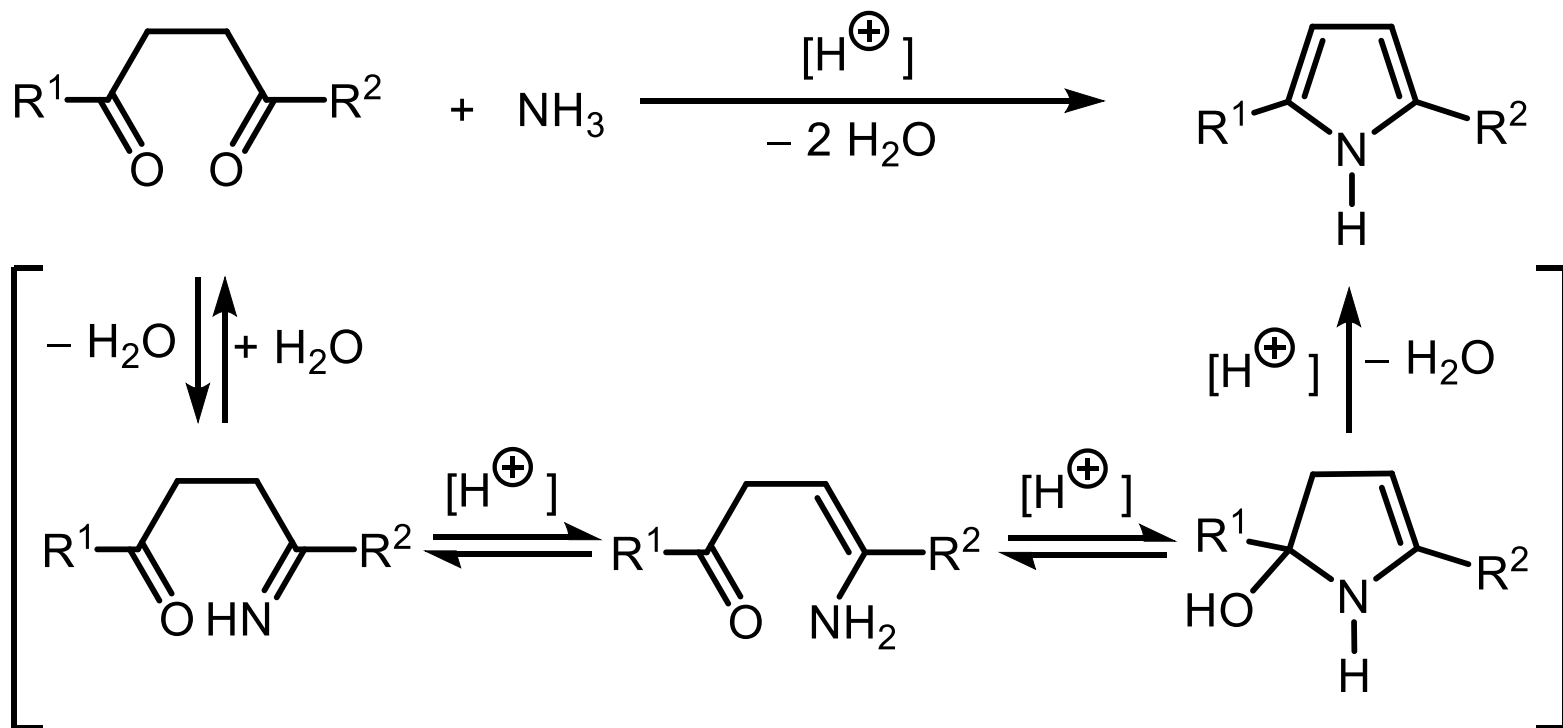
Retrosynthetische Überlegungen



4.4.3.2. Ringaufbauende Synthesen

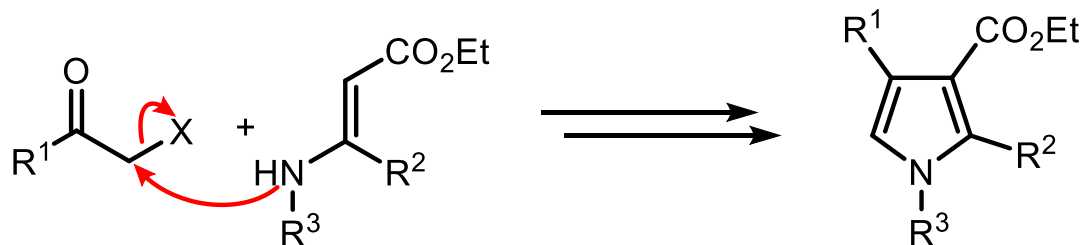
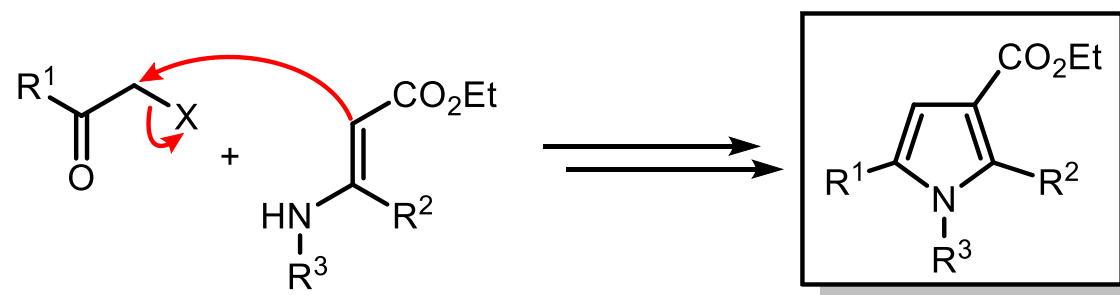
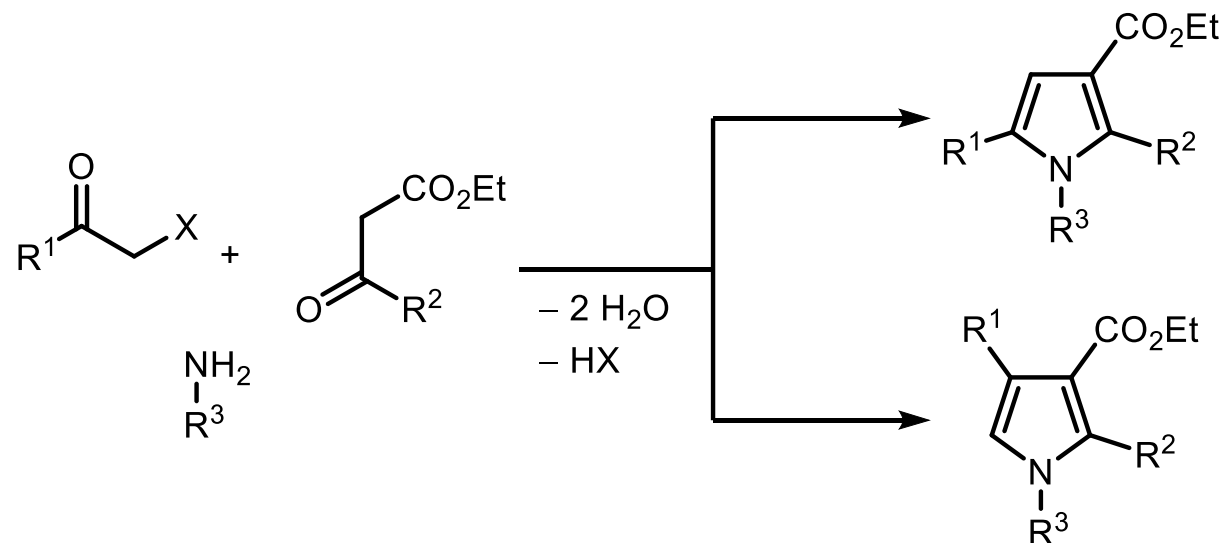
Paal-Knorr-Synthese

52



4.4.3.2. Ringaufbauende Synthesen Hantzsch-Synthese

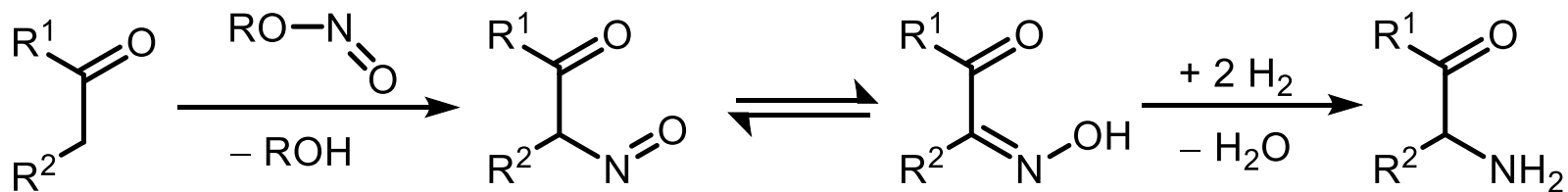
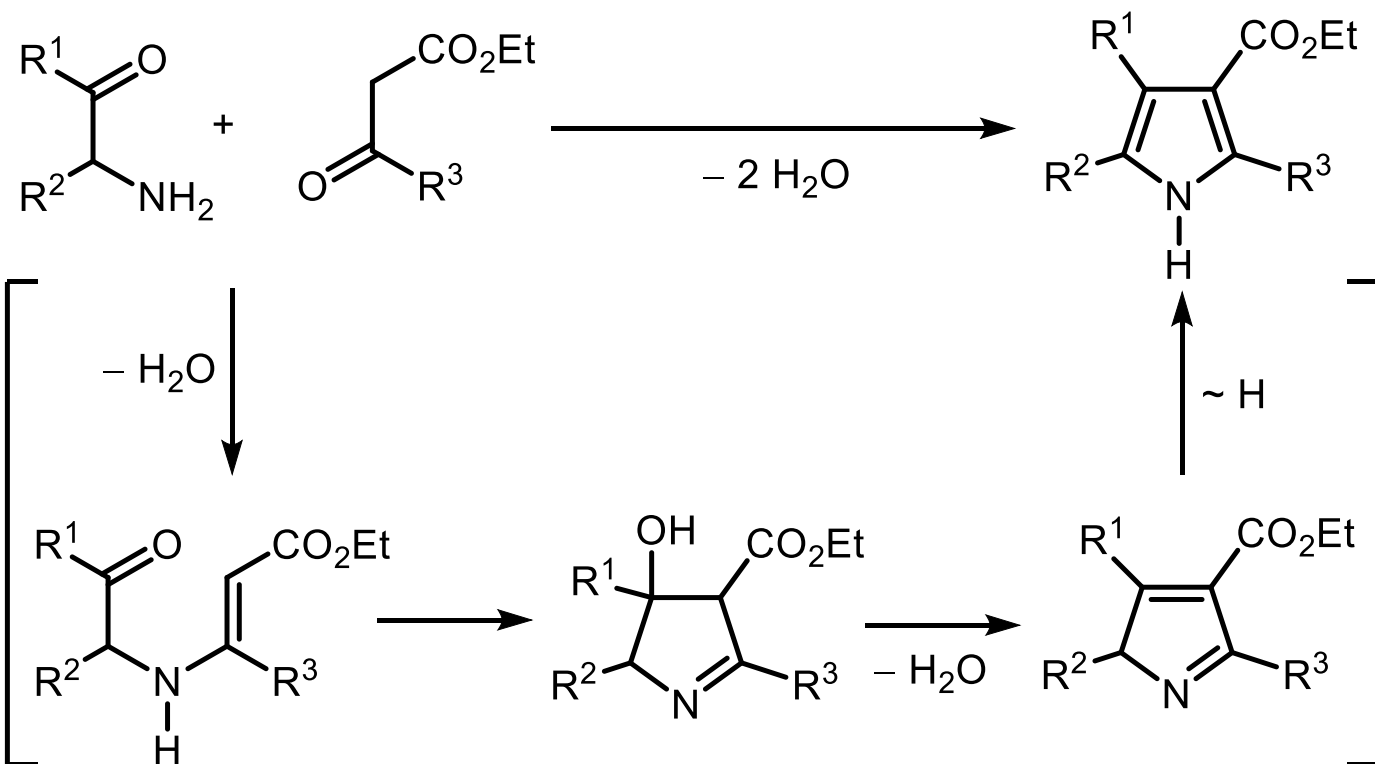
53



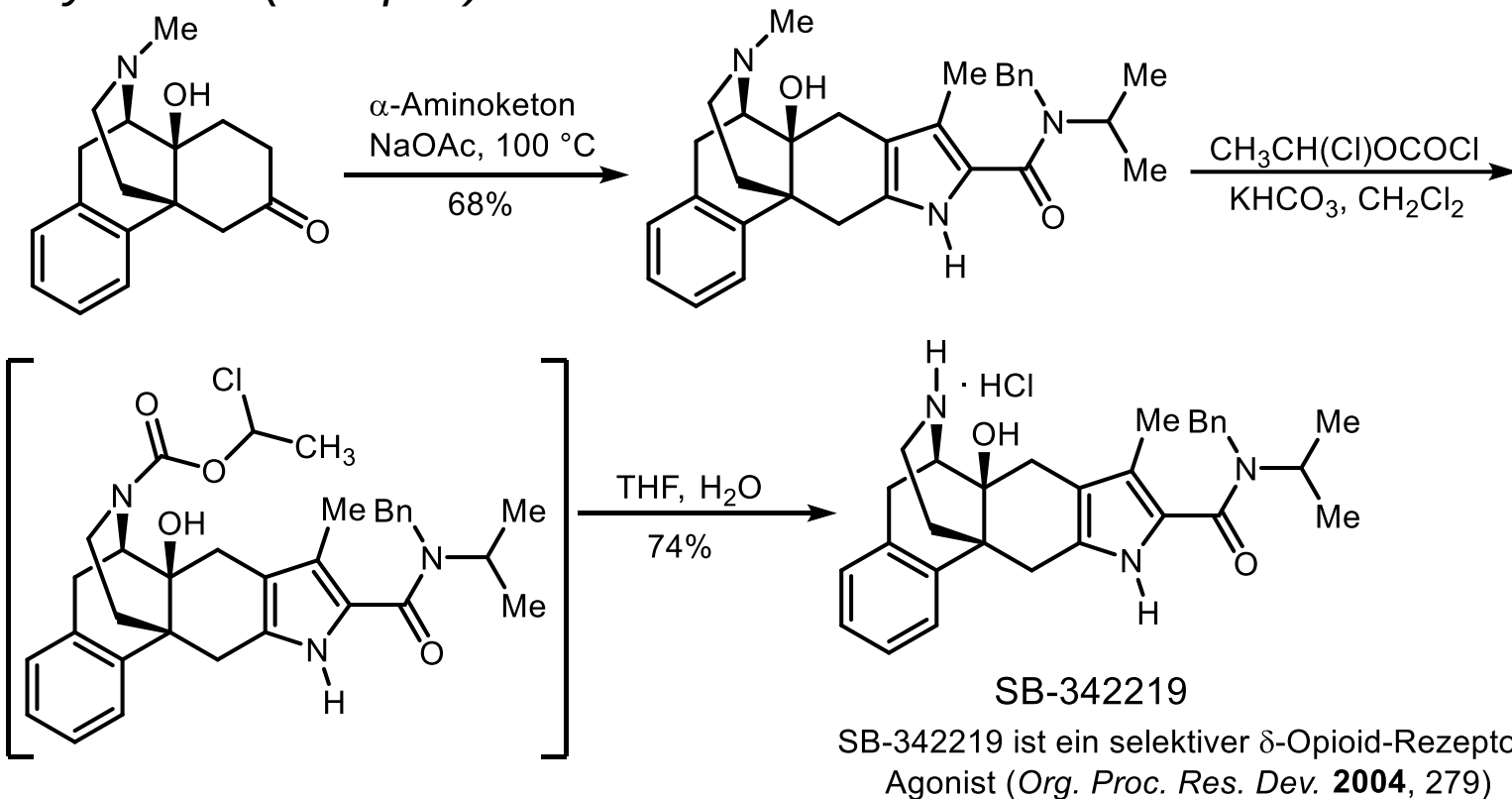
4.4.3.2. Ringaufbauende Synthesen

Knorr-Synthese

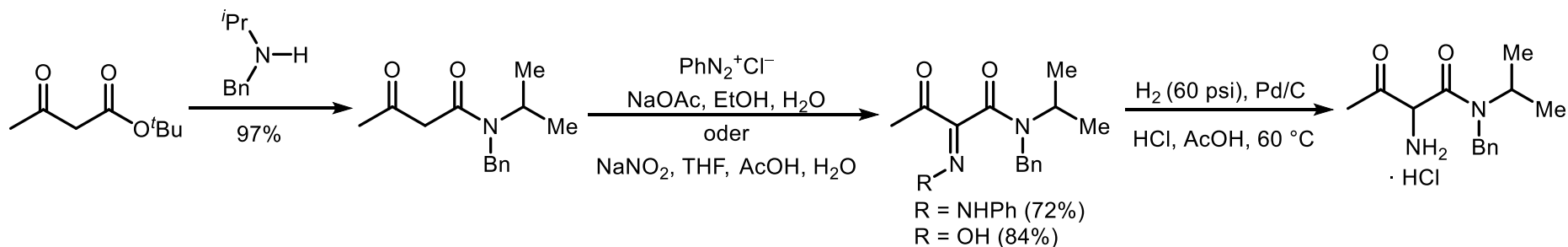
54



4.4.3.2. Ringaufbauende Synthesen Knorr-Synthese (Beispiel)



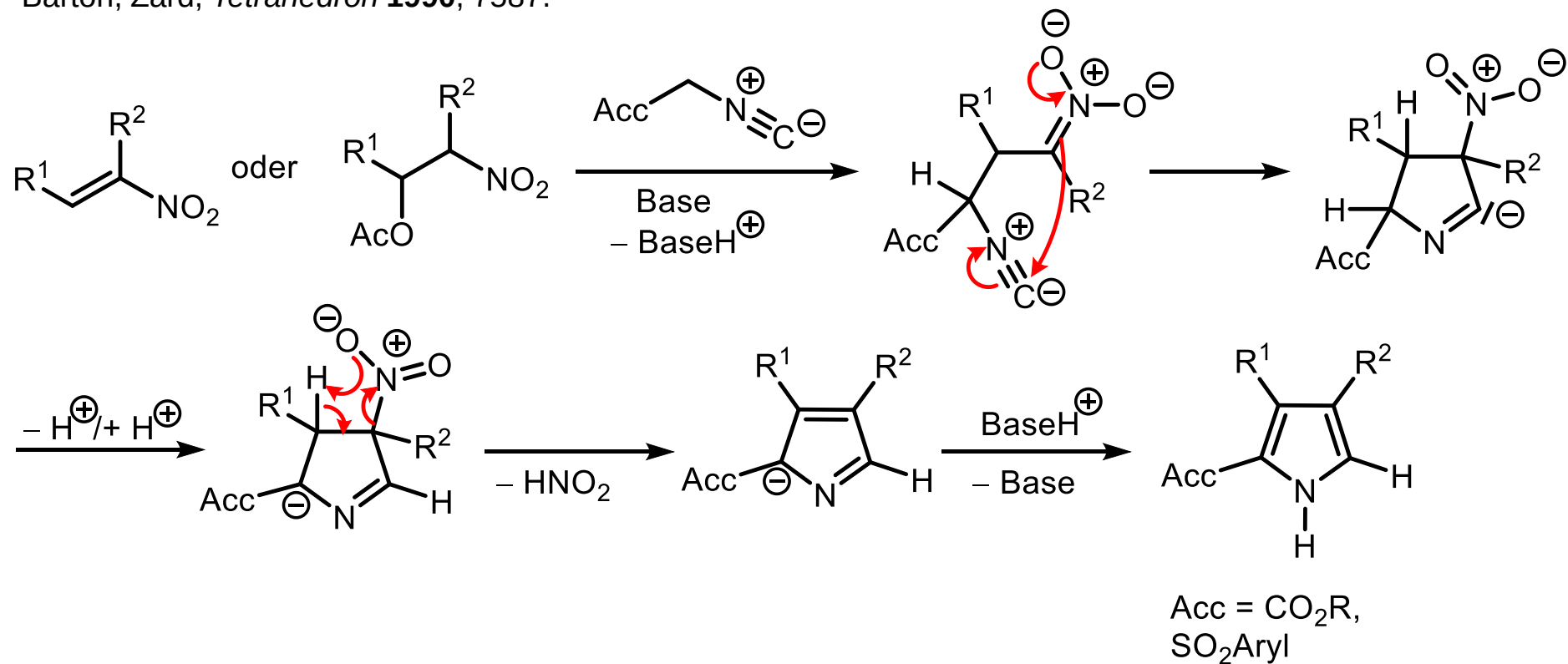
Synthese des α -Aminoketons



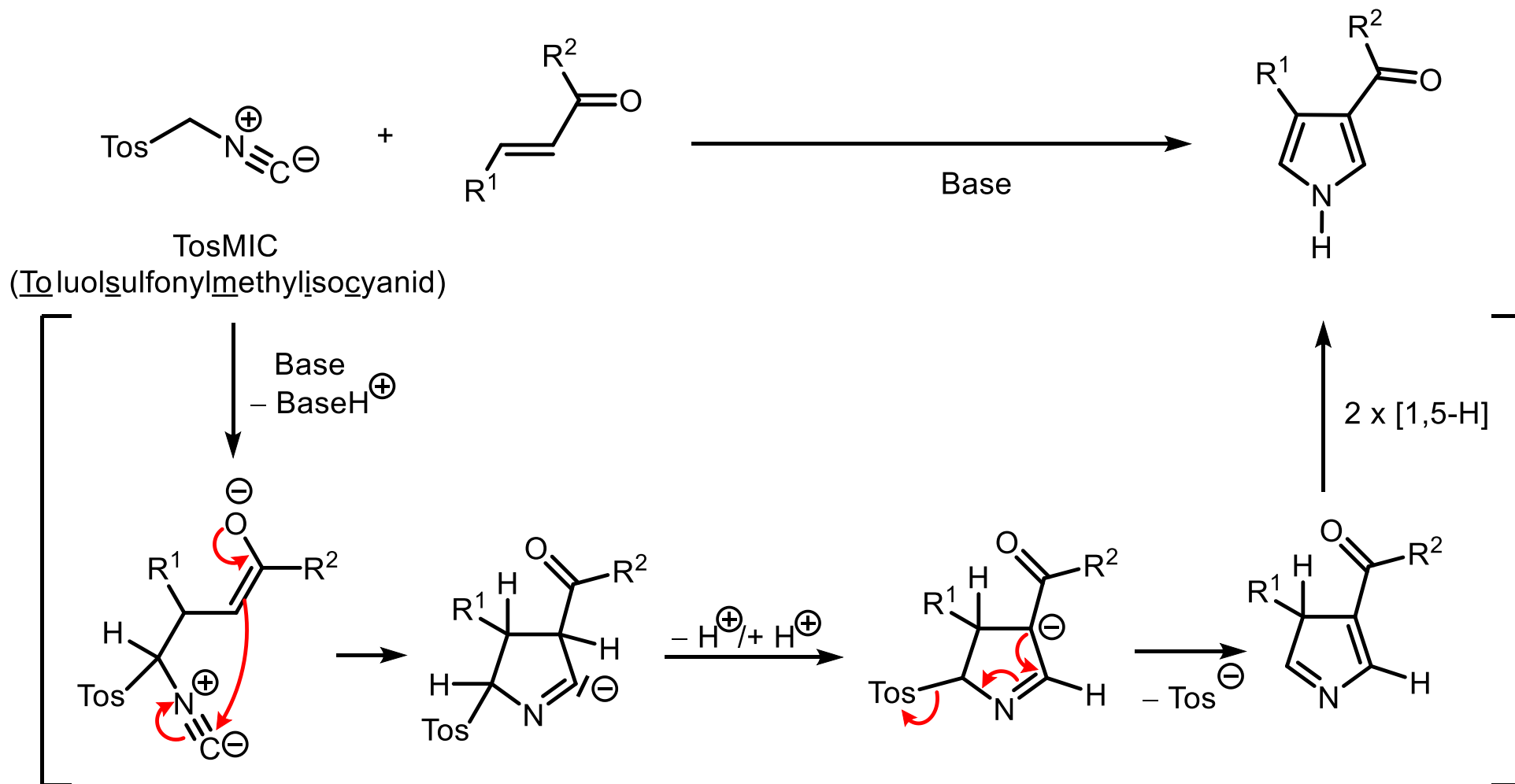
4.4.3.2. Ringaufbauende Synthesen

Barton-Zard-Synthese von α -akzeptor-substituierten Pyrrolen

Barton, Zard, *Tetrahedron* **1990**, 7587.



4.4.3.2. Ringaufbauende Synthesen van-Leusen-Synthese (1972)



4.4.3.2. Ringaufbauende Synthesen Kenner-Synthese

